# Building Energy - Project Report

4. Model Building

I chose Linear Regression for this project. Building energy is pretty straightforward physics -- bigger surface area and taller buildings mean more heat loss which means more heating needed. These relationships are roughly linear so Linear Regression was the natural choice. Plus I can see exactly how much each feature contributes to the final prediction. And since every input was a number I didn't need any encoding at all.

5. Model Evaluation

I used Mean Squared Error and R-squared. The model got an MSE of 4.56 and an $R^2$ of 0.82 -- meaning 82% of the variation in heating load is explained by the five building parameters. That's pretty good. The remaining 18% is probably from stuff I didn't include like insulation quality or climate. The average prediction error was about 2.1 units of heating load which is reasonable for energy estimation.

6. Conclusion

The model predicts building heating load with $R^2$ of 0.82 using five inputs -- surface area, wall area, roof area, height, and glazing. It shows that building geometry has a big impact on energy consumption which matters for designing efficient buildings. The downside is I didn't include insulation, window type, or location/climate data which also affect real energy use. Future improvements would be adding those features and maybe trying Random Forest for comparison.

7. Reflection

This was my first all-numeric project and honestly it was the easiest. No LabelEncoder needed, no figuring out how to convert letters to numbers -- just load, scale, train. I learned that not every ML problem needs a complex model. Sometimes simple Linear Regression is the best tool and you don't need Random Forest or neural networks for everything. The challenge was making sure my generated data made physical sense -- larger buildings needing more heating, taller buildings needing more energy, etc. You can't just throw random numbers together and expect the model to learn anything meaningful. This taught me that domain knowledge matters in ML. The workflow was smooth: load the CSV, scale the features, split the data, train Linear Regression, check $R^2$ and MSE, save model and scaler. I now understand $R^2$ better -- it tells you how much of what you're trying to predict is actually explained by your inputs. 82% is solid but that 18% reminds me that real-world problems always have factors you can't capture.